

BRAC University  
CSE340: Computer Architecture  
Makeup Quiz

Duration : 20 Minutes

Total Marks : 20      Set: Abacus

**ID:**

**Name:**

**Sec:**

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1. Consider the following code:

```
float a = fp_array [7];   The base address of the fp_array array is in x9. The variables d and e are
float b = fp_array [6];   x19 and f19 respectively. The floating point variables are using
float c = fp_array [9];   double-precision.
int d = 78;
float e = (float) d;
if (a >= b) {
    e = a*b;
} else {
    e = b*c;
}
```

- a) Construct the equivalent RISC-V assembly code for the given C program.
- b) Consider  $a = 2.789$  and  $b = 3.892$ . Perform  $a*b$  using IEEE754 double precision floating point representation. Mention if there is any overflow or underflow. Your answer should show all the detailed steps and the final decimal value. Consider 5 digits after the floating point while performing the conversions.
- c) Explain the difference between the following instructions:
- i. `fcvt.s.w f3, x5`
  - ii. `fmv.x.w x3, f5`
- d) Consider the following numbers and different arbitrary-length IEEE 754 floating-point representations. Calculate if the numbers can be represented without any overflow/underflow. Your answer should show the detailed calculations.

| Number                    | Floating point number format             |
|---------------------------|------------------------------------------|
| a. $1.011111011 * 2^{32}$ | Total length: 32 bits, fraction bits: 6  |
| b. $1.00 * 2^{-128}$      | Total length: 32 bits, fraction bits: 18 |
| c. $0.00010110 * 2^{44}$  | Total length: 20 bits, fraction bits: 12 |

